

Quantum Field Theory I Exercise 3

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1 A theory of two scalar fields

The Lagrangian for two massive interacting real scalar fields (with different masses), ϕ and σ , in $3+1$ dimensions, is

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m_\phi^2 \phi^2 + \frac{1}{2}(\partial_\mu \sigma)^2 - \frac{1}{2}m_\sigma^2 \sigma^2 - \frac{\lambda}{2} \phi^2 \sigma \quad (1)$$

1.1 Momentum-space Feynman rules

Let us draw ϕ propegators as solid lines and σ propegators as dashed lines. The rules we get then are:

- Draw fully connected amputated diagrams with n vertices.
- For each ϕ propegator (solid line), add $\frac{i}{p^2 - m_\phi^2 + i\epsilon}$.
- For each σ propegator (dashed line), add $\frac{i}{p^2 - m_\sigma^2 + i\epsilon}$.
- For each vertex add $-i\lambda(2\pi)^4 \delta(p_1 + p_2 + p_3)$.
- Integrate over all undetermined momenta $\int \frac{d^4 p}{(2\pi)^4}$.
- Divide by symmetry factor.
- Remove the remaining $i(2\pi)^4 \delta(\sum p_i)$ factor.

1.2 Coupling constant Scaling dimension

Since

$$[\partial_\mu] = [m_\phi] = [m_\sigma] = 1 \quad (2)$$

And since the Lagrangian has $[\mathcal{L}] = 4$, it's clear to see that:

$$[\phi] = [\sigma] = 1 \quad (3)$$

As well. So, the scaling dimension of the coupling constant is:

$$[\lambda] = 1 \quad (4)$$

This makes this theory a super-renormalizable theory, where the perturbations converge in all energy ranges, but will be more accurate the higher the energy is.

1.3 Calculating $\mathcal{M}$ for $\phi\phi \rightarrow \phi\phi$ scattering

Up to second order in λ , there are three contributing diagrams matching this scattering, one for each of the Mandelstam variables (see figure 1). These diagrams have two vertices, one σ propagator, and no symmetry factor ($S = 1$). This gives:

$$\begin{aligned} & -\lambda^2 \int \frac{d^4 q}{(2\pi)^4} (2\pi)^4 \delta(p_1 + p_2 - q) (2\pi)^4 \delta(q - p_3 - p_4) \frac{i}{q^2 - m_\sigma^2 + i\epsilon} + \\ & \quad + (p_2 \leftrightarrow -p_3) + (p_2 \leftrightarrow -p_4) + O(\lambda^3) = \\ & = -i\lambda^2 (2\pi)^4 \delta(p_1 + p_2 - p_3 - p_4) \frac{1}{(p_1 + p_2)^2 - m_\sigma^2 + i\epsilon} + \\ & \quad + (p_2 \leftrightarrow -p_3) + (p_2 \leftrightarrow -p_4) + O(\lambda^3) \end{aligned} \quad (5)$$

And so

$$\mathcal{M} = -\lambda^2 \left[\frac{1}{s^2 - m_\sigma^2 + i\epsilon} + \frac{1}{t^2 - m_\sigma^2 + i\epsilon} + \frac{1}{u^2 - m_\sigma^2 + i\epsilon} \right] + O(\lambda^3) \quad (6)$$

1.4 Calculating $\mathcal{M}$ for $\phi\phi \rightarrow \sigma\sigma$ scattering

This time, the only contributing diagrams correspond to the t and u channel, with the s channel not applying (see figure 2). The result is otherwise the same - 2-vertices 1-propagator diagrams, only this time the propagator is a ϕ propagator:

$$\mathcal{M} = -\lambda^2 \left[\frac{1}{t^2 - m_\phi^2 + i\epsilon} + \frac{1}{u^2 - m_\phi^2 + i\epsilon} \right] + O(\lambda^3) \quad (7)$$

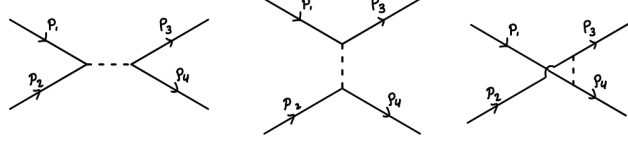


Figure 1: The three diagrams matching the three Mandelstam variables, from right to left, s , t and u . The internal propagator is a σ propagator.

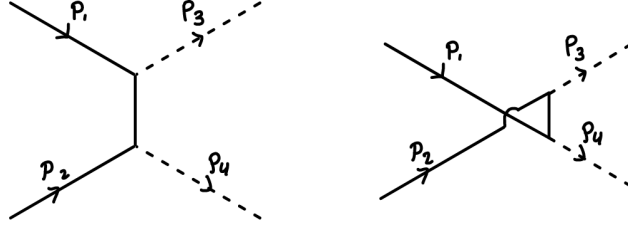


Figure 2: The two diagrams contributing to the $\phi\phi \rightarrow \sigma\sigma$ process.

1.5 The $\sigma\sigma \rightarrow \sigma\sigma$ scattering

For the $\sigma\sigma \rightarrow \sigma\sigma$ scattering to occur, at least four vertices are needed. This is because any interaction of the incoming σ fields can only result in ϕ fields, which must then again interact to create σ fields. Since there is only one σ field in each vertex, this requires one vertex for each of the incoming fields and one vertex for each of the outgoing fields. The most simple diagram accomplishing this is the four point loop diagram:

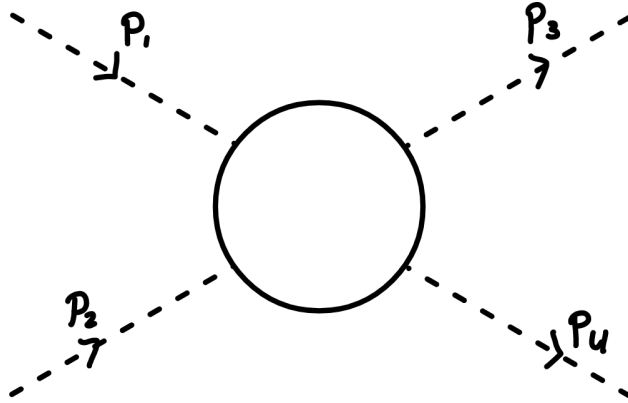


Figure 3: The four point diagram loop for the $\sigma\sigma \rightarrow \sigma\sigma$ scattering.

It has four vertices and four internal ϕ propagators, and since it has a loop

that can swithc direction it has a dymmetry factor $S = 2$. This gives:

$$\begin{aligned}
& \lambda^4 \int \prod_{i=1}^4 \frac{d_i^q}{(2\pi)^4} (2\pi)^4 \delta(p_1 + q_1 - q_2) (2\pi)^4 \delta(p_2 + q_2 - q_4) \times \\
& \times (2\pi)^4 \delta(p_3 - q_1 - p_3) (2\pi)^4 \delta(q_4 - q_3 - p_4) \prod_{i=0}^4 \frac{i}{q_i^2 - m_\phi^2 + i\epsilon} = \\
& = \lambda^4 \int \frac{d^q}{(2\pi)^4} (2\pi)^4 \delta(p_1 + p_2 - p_3 - p_4) \frac{1}{q^2 - m_\phi^2 + i\epsilon} \times \\
& \times \frac{1}{(q + p_1)^2 - m_\phi^2 + i\epsilon} \frac{1}{(q + p_3)^2 - m_\phi^2 + i\epsilon} \frac{1}{(q + p_3 + p_4)^2 - m_\phi^2 + i\epsilon} =
\end{aligned} \tag{8}$$

Where we first used the δ functions to integrate three of the four q_i s, and got:

$$\begin{aligned}
q_3 &= q_1 + p_1 & q_4 &= q_1 + p_3 + p_4 & q_2 &= q_1 + p_3 + p_4 - p_2 \\
q_1 &= q_1 + p_3 + p_4 - p_2 - p_1 \rightarrow p_1 + p_2 & &= p_3 + p_4
\end{aligned} \tag{9}$$

Then, we names $q_1 = q$ and left it as a last integratin variable, noticing that $q_2 = q_1 + p_1$. And so the contribution of this diagram to the scattering amplitude is:

$$\begin{aligned}
\mathcal{M} &= \lambda^4 \int \frac{d^q}{(2\pi)^4} \frac{1}{q^2 - m_\phi^2 + i\epsilon} \times \\
& \times \frac{1}{(q + p_1)^2 - m_\phi^2 + i\epsilon} \frac{1}{(q + p_3)^2 - m_\phi^2 + i\epsilon} \frac{1}{(q + p_3 + p_4)^2 - m_\phi^2 + i\epsilon}
\end{aligned} \tag{10}$$

This diagram has $L = 1$ loops, and $P = 4$ internal propegators, and so the superficial degree of divergence is

$$D = d \cdot L - 2P = 4 \cdot 1 - 8 = -4 < 0 \tag{11}$$

So we got a negative superficial degree of freedom, and since there are no subdia-grams, the integral converges.

1.6 The symmetry of the Lagrangian

The given Lagrangian has a discrete $\mathbb{Z}_2$ symmetry

$$\phi, \sigma \rightarrow -\phi, \sigma \tag{12}$$

This means that for $\phi\phi \rightarrow \phi\sigma$ scattering, the initial and final states have different parity, and since the symmetry gives parity preservation, the transformation is forbidden.

2 Different regularizations for the Casimir force in one spatial dimension

2.1 An expression for the Casimir force

Since the vacuum energy of each part of the system is:

$$\langle \Omega | H(\Delta x) | \Omega \rangle = \langle \Omega | \sum_{n=1}^{\infty} \omega_n(\Delta x) \left(a_n^\dagger a_n + \frac{1}{2} \right) | \Omega \rangle = \frac{1}{2} \sum_{n=1}^{\infty} \omega_n(\Delta x) \quad (13)$$

Where

$$\Delta x = \frac{\pi n}{\Delta x} \quad (14)$$

This becomes:

$$\langle \Omega | H(\Delta x) | \Omega \rangle = \frac{\pi}{2\Delta x} \sum_{n=1}^{\infty} n \quad (15)$$

The total vacuum energy of both systems is then:

$$\begin{aligned} E_{tot}(l) &= \langle \Omega | H(l) | \Omega \rangle + \langle \Omega | H(R-l) | \Omega \rangle = \frac{\pi}{2} \left(\sum_{n=1}^{\infty} n \right) \left(\frac{1}{l} + \frac{1}{R-l} \right) = \\ &= \frac{\pi}{2} \frac{R}{l(R-l)} \left(\sum_{n=1}^{\infty} n \right) \end{aligned} \quad (16)$$

The force is then given by the derivative with respect to l :

$$F_c = -\frac{dE_{tot}}{dl} = \frac{\pi}{2} \frac{R(R-2l)}{l^2(R-l)^2} \left(\sum_{n=1}^{\infty} n \right) \quad (17)$$

Which, even taking $R \rightarrow \infty$, diverges because of the sum:

$$\lim_{R \rightarrow \infty} F_c = \frac{\pi}{2l^2} \left(\sum_{n=1}^{\infty} n \right) \rightarrow \infty \quad (18)$$

2.2 Heat-kernel regularization

Let us use the regularization

$$\omega_n \rightarrow \omega_n \exp\left[-\frac{\omega_n}{\pi\Lambda}\right] = \frac{\pi n}{\Delta x} \exp\left[-\frac{n}{\Delta x\Lambda}\right] \quad (19)$$

For some large Λ . This gives:

$$E_{tot}(l) = \frac{\pi}{2} \left[\frac{1}{l} \left(\sum_{n=1}^{\infty} n e^{-n/\Lambda l} \right) + \frac{1}{R-l} \left(\sum_{n=1}^{\infty} n e^{-n/\Lambda(R-l)} \right) \right] \quad (20)$$

Now, if we mark $y = \Lambda^{-1}$, we get the energy inside the box:

$$\begin{aligned} \frac{1}{l} \left(\sum_{n=1}^{\infty} n e^{-n/\Lambda l} \right) &= \sum_{n=1}^{\infty} \frac{n}{l} e^{-ny/l} = -\frac{\partial}{\partial y} \sum_{n=1}^{\infty} e^{-ny/l} = -\frac{\partial}{\partial y} \frac{1}{e^{y/l} - 1} = \\ &= -\frac{\partial}{\partial y} \left(\frac{l}{y} - \frac{1}{2} + \frac{y}{12l} - \frac{y^3}{720l^3} + \dots \right) = \frac{l}{y^2} - \frac{1}{12l} + \frac{y^2}{240l^3} + \dots = \\ &= \frac{l}{y^2} - \frac{1}{12l} + \frac{y^2}{240l^3} + \dots \approx l\Lambda^2 - \frac{1}{12l} \end{aligned} \quad (21)$$

Where the the two step we neglected all negative orders in Λ . Acting similarly on the energy for the states outside the box, we get that the total energy is:

$$\begin{aligned} E_{tot}(l) &= \frac{\pi}{2} \left(l\Lambda^2 - \frac{1}{12l} + (R-l)\Lambda^2 - \frac{1}{12(R-l)} \right) = \\ &= \frac{\pi}{2} \left(R\Lambda^2 - \frac{1}{12} \frac{R}{l(R-l)} \right) \end{aligned} \quad (22)$$

For large R , the first order contribution is $\frac{\pi}{2} R\Lambda^2$. And the Casimir force:

$$F_c = -\frac{dE_{tot}}{dl} = \frac{\pi}{24} \frac{R(2l-R)}{l^2(l-R)^2} \xrightarrow{R \rightarrow \infty} -\frac{\pi}{24l^2} \quad (23)$$

2.3 ζ function regularization

Let us use the regularization

$$\omega_n \rightarrow \omega_n \left(\frac{\omega_n}{\mu} \right)^{-s} = \left(\frac{\pi}{\Delta x} \right)^{1-s} \mu^s n^{1-s} \quad (24)$$

Where we will take $s \rightarrow 0$ at the end. This gives the energy inside the box:

$$\left(\frac{\pi}{l} \right)^{1-s} \mu^s \left(\sum_{n=1}^{\infty} n^{1-s} \right) = \left(\frac{\pi}{l} \right)^{1-s} \mu^s \zeta(s-1) \xrightarrow{s \rightarrow 0} -\frac{\pi}{12l} \quad (25)$$

This gives the total energy:

$$E_{tot}(l) = -\frac{\pi}{24} \frac{R}{l(R-l)} \quad (26)$$

Which has the same l dependency as in the Heat kernel regularization, giving the same force:

$$F_c = -\frac{dE_{tot}}{dl} = -\frac{\pi}{24} \frac{R(2l-R)}{l^2(R-l)^2} \xrightarrow{R \rightarrow \infty} -\frac{\pi}{24l^2} \quad (27)$$

2.4 General regularization

For a general regularization

$$\omega_n \rightarrow \omega_n f\left(\frac{\omega_n}{\Lambda}\right) \quad (28)$$

With a function $f(x)$ that has the properties

$$\lim_{x \rightarrow \infty} x f(x) = 0 \quad f(0) = 1 \quad (29)$$

The total energy becomes:

$$E_{tot}(l) = \frac{\pi}{2} \left[\sum_{n=1}^{\infty} \frac{n}{l} f\left(\frac{\pi n}{l\Lambda}\right) + \sum_{n=1}^{\infty} \frac{n}{R-l} f\left(\frac{\pi n}{(R-l)\Lambda}\right) \right] \quad (30)$$

Using the hint, let us take the large R limit to approximate the second sum to an integral. Taking $x = \frac{n}{(R-l)\Lambda}$:

$$\begin{aligned} & \sum_{n=1}^{\infty} \frac{n}{R-l} f\left(\frac{\pi n}{(R-l)\Lambda}\right) = \\ &= \frac{\Lambda^2(R-l)}{\pi^2} \sum_{n=1}^{\infty} \frac{\pi n}{(R-l)\Lambda} f\left(\frac{\pi n}{(R-l)\Lambda}\right) \frac{\pi}{\Lambda(R-l)} \approx \\ &\approx \frac{\Lambda^2(R-l)}{\pi^2} \int_0^{\infty} x f(x) dx \equiv \frac{\Lambda^2(R-l)}{\pi^2} I \end{aligned} \quad (31)$$

Where we defined I as the numeric value from the intergration over $x f(x)$. Now let us turn to the first sum. We will use the Euler-Maclaurin formula for the difference between a sum from a to b to an integral on $[a, b]$:

$$\begin{aligned} & \sum_{n=a}^b f(n) - \int_a^b f(x) dx = \\ &= \frac{1}{2} (f(b) + f(a)) + \frac{1}{12} (f'(b) - f'(a)) - \frac{1}{30} (f'''(b) - f'''(a)) + \dots \end{aligned} \quad (32)$$

Before we apply it, we must do a change of variables. Let us rewrite the first sum:

$$\sum_{n=1}^{\infty} \frac{n}{l} f\left(\frac{\pi n}{l\Lambda}\right) = \frac{\Lambda}{\pi} \sum_{n=1}^{\infty} \frac{\pi n}{l\Lambda} f\left(\frac{\pi n}{l\Lambda}\right) = \frac{\Lambda}{\pi} \sum_{n=1}^{\infty} x f(x) \quad (33)$$

Where we redefined x in terms of l instead of $(R-l)$. Since the derivatives in the formula are based on the summation variable n , each derivative adds another factor of Λ^{-1} . Since there is a Λ^1 outside the sum, all derivatives of order two or higher will have negative powers of Λ , and we can ignore them. This leaves only the first two terms on the RHS:

$$\sum_{n=a}^b f(n) - \int_a^b f(x) dx = \frac{1}{2} (f(b) + f(a)) + \frac{1}{12} (f'(b) - f'(a)) \quad (34)$$

Let us now define:

$$g(x) = g\left(\frac{\pi n}{l\Lambda}\right), \quad g(x) = x f(x) \quad (35)$$

And derive:

$$\frac{dg}{dn} = \frac{dg}{dx} \frac{dx}{dn} = \frac{\pi}{l\Lambda} [f(x) + x f'(x)] \quad (36)$$

Finally, let us notice that:

$$\begin{aligned} g(0) &= 0 \cdot f(0) = 0 \cdot 1 = 0 & \lim_{x \rightarrow \infty} g(x) &= 0 \\ \frac{dg}{dn} \Big|_0 &= \frac{\pi}{l\Lambda} [f(0) + 0 \cdot f'(0)] = \frac{\pi}{l\Lambda} \\ \lim_{x \rightarrow \infty} \frac{dg}{dn} \Big|_x &= \lim_{x \rightarrow \infty} \frac{\pi}{l\Lambda} [f(x) + x f'(x)] = \frac{\pi}{l\Lambda} \lim_{x \rightarrow \infty} x f'(x) = \frac{\pi^2}{l^2 \Lambda^2} \lim_{n \rightarrow \infty} n f'\left(\frac{\pi n}{l\Lambda}\right) \end{aligned} \quad (37)$$

The last term is again of order Λ^{-2} and thus we can ignore it. Pluggin this into the above equation with $a=0$ and $b=\infty$, and using the Jacobean $dn = \frac{\Lambda l}{\pi}$:

$$\sum_{n=1}^{\infty} g(n) - \frac{\Lambda l}{\pi} I = \frac{1}{2} \left(g(0) + \lim_{x \rightarrow \infty} g(x) \right) + \frac{1}{12} \left(\lim_{x \rightarrow \infty} \frac{dg}{dn} \Big|_x - \frac{dg}{dn} \Big|_0 \right) = -\frac{\pi}{12 l \Lambda} \quad (38)$$

The total energy is then:

$$\begin{aligned} E_{tot}(l) &= \frac{\pi}{2} \left[\sum_{n=1}^{\infty} \frac{n}{l} f\left(\frac{\pi n}{l\Lambda}\right) + \sum_{n=1}^{\infty} \frac{n}{R-l} f\left(\frac{\pi n}{(R-l)\Lambda}\right) \right] = \\ &= \frac{\pi}{2} \left[\frac{\Lambda}{\pi} \left(\frac{\Lambda l}{\pi} I - \frac{\pi}{12 l \Lambda} \right) + \frac{\Lambda^2 (R-l)}{\pi^2} I \right] = \\ &= -\frac{\pi}{24 l} + \frac{\Lambda^2 R}{2 \pi} I \end{aligned} \quad (39)$$

And the force:

$$F_c = -\frac{dE_{tot}}{dl} = -\frac{\pi}{24 l^2} \quad (40)$$

As expected.